Summary in Graph

Exam Summary (GO Classes GATE CS | Data Structures | Weekly Quiz 5)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	10 (8 + 2)
Total Marks:	12 (8 + 4)
Exam Duration:	30 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained

Q1 Q2 Q3 Q4 Q5 Q6 Q7 Q8 Q9 Q10

* Average Stats based on first-attempt stats of Top 10% (19) out of 188 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 Data Structures

We found a new implementation of a priority queue where insert takes constant time and extractMin takes $O(\log n)$ time. Which of the following is true?

A. This can be used to sort in constant time

B. This can be used to sort in linear time

C. This can be used to sort in $O(n \log n)$ time

D. For sorting, one needs a hash table, not a priority queue

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 02sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Consider a binary search tree whose labels are each one capital letter, and assume that a level-order traversal yields the sequence "GENIUS". What is the preorder traversal of this tree?

A. GENIUS

B. GNEUIS

C. GUINES

D. GIUSEN

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Select Type | Award: 1 | Penalty: 0 | Data Structures

Consider a sorted circular doubly linked list where the head element points to the smallest element in the list.

Which of the following is/are true?

A. $O(1)$ is the asymptotic complexity of finding the smallest element in the list

B. $O(1)$ is the asymptotic complexity of finding the largest element in the list

C. $O(n)$ is the asymptotic complexity of determining whether a given element e appears in the list

D. $O(1)$ is the asymptotic complexity of deleting a given element e in the list (not including the cost of finding it)

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Consider a program P that does not use recursion. P is a program which tries to reverse a queue using other auxiliary queues.

Which of the following is true about P?

A. P reverses a given queue using a minimum of one auxiliary queue.

B. P reverses a given queue using a minimum of two auxiliary queues.

C. P reverses a given queue using a minimum of three auxiliary queues.

D. P reverses a given queue using a minimum of four auxiliary queues.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Suppose that you implement a queue using a null-terminated singly linked list, maintaining a reference to the item least recently added (the front of the list) but not maintaining a reference to the item most recently added (the end of the list).
What are the worst-case running times for enqueue and dequeue, respectively?

A. $O(1)$ for both operations

B. $O(1)$ for enqueue and $O(n)$ for dequeue

C. $O(n)$ for enqueue and $O(1)$ for dequeue

D. $O(n)$ for both operations

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Consider two statements $\mathrm{S1}$ and $\mathrm{S2}$ below.

- $\mathrm{S1}$: If $f_1(n) = O\left(g_1(n)\right)$ and $f_2(n) = O\left(g_2(n)\right)$, then $f_1(n) + f_2(n) = O\left(\max\left\{g_1(n), g_2(n)\right\}\right)$.
- $\mathrm{S2}$: If $f_1(n) = \Omega\left(g_1(n)\right)$ and $f_2(n) = \Omega\left(g_2(n)\right)$, then $f_1(n) + f_2(n) = \Omega\left(\min\left\{g_1(n), g_2(n)\right\}\right)$.

Which of the following is correct?

A. $\mathrm{S1}$ is true but $\mathrm{S2}$ is false

B. $\mathrm{S1}$ is false but $\mathrm{S2}$ is true

C. Both statements are true

D. Both statements are false

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Consider two statements, $\mathrm{S1}$ and $\mathrm{S2}$, below.

- $\mathrm{S1}$: An $\Theta(n \log n)$ algorithm ALWAYS runs faster than an $\Theta\left(n^2\right)$ algorithm.
- $\mathrm{S2}$: An $\Theta(n \log n)$ algorithm ALWAYS runs slower than an $\Theta\left(n^2\right)$ algorithm.

Which of the following is correct?

A. $\mathrm{S1}$ is true, but $\mathrm{S2}$ is false

B. $\mathrm{S1}$ is false, but $\mathrm{S2}$ is true

C. Both statements are true

D. Both Statements are false

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Let $f(n) = \Omega(n), g(n) = O(n)$ and $h(n) = \Theta(n)$. Then $[f(n).\, g(n)] + h(n)$ is:

A. $\Omega(\mathrm{n})$

B. $O(n)$

C. $\theta(n)$

D. None of these

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

Consider 2 problems:

1. Problem 1 : Given a queue Q containing n elements, transfer these items on to a stack S (initially empty) so that the front element of Q appears at the top of the stack and the order of all other items is preserved. Using enqueue and dequeue operations for the queue, and push and pop operations for the stack.
2. Problem 2 : Generate the permutations possible when using only a single stack and push and pop are the only operations allowed.

- $\mathrm{S1}$: An efficient algorithm to solve problem 1 will take $O(n)$ time and a constant amount of additional storage.
- $\mathrm{S2}$: The no of stack permutations on a stack with n elements is equal to the number of balanced parentheses with $2n$ total parentheses.

Which of the following is correct?

A. Only $\mathrm{S1}$ is correct
B. Only $\mathrm{S2}$ is correct
C. Both $\mathrm{S1}$ and $\mathrm{S2}$ are correct
D. None of them is correct

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Select Type | Award: 2 | Penalty: 0 | Data Structures

An SQueue is a queue implemented using two stacks. SQueue supports two operations enqueue() and dequeue().
Consider a below pseudo code that implements enqueue() and dequeue() for SQueue using two stacks inbox and outbox.

```
    Stack inbox, outbox;
    void enqueue(int item) {
        inbox.push(item);
    }
void dequeue() {
    if (outbox.isEmpty())
    while (!inbox.isEmpty())
    outbox.push(in.pop());
    return outbox.pop();
}
```

Consider the state of SQueue which is initialized with 100 items i.e. 100 enqueue operations have been performed in SQueue.

Which of the following(s) is/are true if we want to dequeue all elements from SQueue?

A. First dequeue() will result in a total of 200 calls to push and pop operations.

B. Apart from the first dequeue(), each of the remaining calls to dequeue will take a total of 100 calls to push and pop operations.

C. Total calls to push and pop were required to dequeue all 100 elements is 300.

D. Total calls to push and pop were required to dequeue all 100 elements is 301.

Your Answer: **Correct Answer: C** **Not Attempted** **Time taken: 00min 00sec** **Discuss**

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